

STATISTICS FOR BUSINESS ASSIGNMENT – 2

Term Deposit Subscription Prediction

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1.0 Introduction and Background

Understanding customer response to marketing campaigns is a key challenge for financial institutions, given high acquisition costs and limited marketing resources. In banking, term deposit subscriptions provide stable funding, and improving direct marketing effectiveness can enhance profitability and efficiency. Prior research shows that customer demographics, contact strategies, and macroeconomic conditions influence subscription decisions (Moro, Cortez and Rita, 2014; Lessmann et al., 2015). Direct marketing has been studied extensively due to its structured, measurable outcomes. Variables such as contact duration, frequency, and previous campaign results are strong predictors of response (Santos, Cortez and Rita, 2018), while socio-demographic factors like age, education, and marital status, and macroeconomic indicators such as employment and interest rates, affect willingness to invest in fixed-term products (Beckmann and Menkhoff, 2008; Mishkin, 2016). This study analyses a bank marketing dataset with demographic, campaign, and macroeconomic variables, using logistic regression to estimate subscription probabilities (Gujarati, 2020). Based on existing literature and business reasoning, the following hypotheses were developed:

- H1: Customer demographic characteristics significantly influence the likelihood of subscription.
- H2: Marketing campaign characteristics significantly affect subscription probability.
- H3: Positive outcomes in previous campaigns increase the likelihood of subscription.
- H4: Macroeconomic indicators significantly influence subscription decisions.
- H5: A multivariate logistic regression model provides superior predictive performance compared to simpler models.

These hypotheses guide the analysis and support the development of predictive models aimed at improving data-driven marketing decisions in the banking sector. Previous studies have benchmarked various classification algorithms, demonstrating that multivariate approaches outperform simpler models in predicting customer subscription (Lessmann et al., 2015)

2.0 Methodology

This study employs a structured, hypothesis-driven approach to identify determinants of customer subscription to term deposits, following established practices in business analytics

and marketing response modelling to ensure statistical validity and reproducibility (Gujarati, 2020). All analyses were conducted using R, with code provided in the appendix.

2.1 Data Preparation and Quality Assessment

The dataset includes customer demographics, marketing campaign characteristics, and macroeconomic indicators, with a binary outcome indicating subscription status. Initial screening involved checking for missing values, duplicates, and data type inconsistencies. Duplicate rows were removed, and categorical variables—such as gender, marital status, education level, contact method, and previous campaign outcome—were converted to factors. The dependent variable (“subscribed”) was coded as a binary factor (“no”, “yes”) for logistic regression. Outliers were identified using boxplots but retained, as extreme values may reflect genuine customer behaviour (Wooldridge, 2015).

2.2 Exploratory Data Analysis

Exploratory analysis examined variable distributions and preliminary relationships with subscription outcomes using summary statistics, frequency tables, and visualisations such as boxplots, bar charts, and density plots. These analyses informed hypothesis testing and guided predictor selection for regression modelling, a key step to ensure models capture relevant customer behavior patterns (Tan, Steinbach and Kumar, 2019).

2.3 Measures of Association and Hypothesis Testing

Associations were assessed using chi-square tests for categorical variables and t-tests for continuous variables. Correlation analysis evaluated relationships among numeric predictors and identified potential multicollinearity, justifying variables’ inclusion in multivariate models.

2.4 Model Development and Evaluation

Logistic regression was the primary technique due to the binary outcome (Hosmer, Lemeshow and Sturdivant, 2013). The data were split into training (70%) and test (30%) sets with stratified sampling. Three nested models were estimated: demographic-only, extended with marketing variables, and full including macroeconomic indicators. Model fit was compared via AIC. Predictive performance was assessed using confusion matrices, ROC curves, and AUC (Fawcett, 2006), with reproducibility facilitated by R (Kuhn and Johnson, 2013). Alternative methods like support vector machines have been applied in similar tasks (Cortes and Vapnik, 1995; Bell, Koren and Volinsky, 2007).

2.5 Ethical Considerations

The dataset contained no personally identifiable information. All analytical decisions were documented transparently, ensuring ethical and reproducible research practices.

3.0 Results and Discussion

3.1 Descriptive Statistics and Data Characteristics

desc_stats							
	Variable	Mean	Median	SD	Min	Max	
age	age	NA	NA	NA	NA	NA	
contact_duration	contact_duration	258.45508470	180.000	259.5580016	0.000	4918.000	
number_contacts	number_contacts	2.56551775	2.000	2.7704816	1.000	56.000	
last_contacted	last_contacted	962.27752282	999.000	187.3972896	0.000	999.000	
prev_contacts	prev_contacts	0.17387590	0.000	0.4960600	0.000	7.000	
empl_var_rate	empl_var_rate	0.07492555	1.100	1.5722117	-3.400	1.400	
cons_price_index	cons_price_index	93.57628099	93.798	0.5802366	92.201	94.767	
cons_conf_index	cons_conf_index	-40.52538202	-41.800	4.6286490	-50.800	-26.900	
euribor_three_mth	euribor_three_mth	3.61430808	4.857	1.7362159	0.634	5.045	
nr_employed	nr_employed	5166.72436655	5191.000	72.2876963	4963.600	5228.100	

Descriptive statistics indicate substantial heterogeneity across key campaign and economic variables. The average contact duration is approximately 258 seconds (SD = 259.56), with values ranging from 0 to 4,918 seconds, highlighting wide variation in customer engagement. The mean number of contacts per customer is 2.57, suggesting that most individuals were contacted only a small number of times, consistent with typical direct marketing strategies.

Macroeconomic indicators exhibit comparatively lower dispersion. The mean Euribor three-month rate is 3.61, while the average number of employed individuals is approximately 5,167, reflecting relatively stable economic conditions during the campaign period. Overall, the descriptive statistics suggest a dataset that captures both behavioural variability and macroeconomic context relevant to subscription decisions.

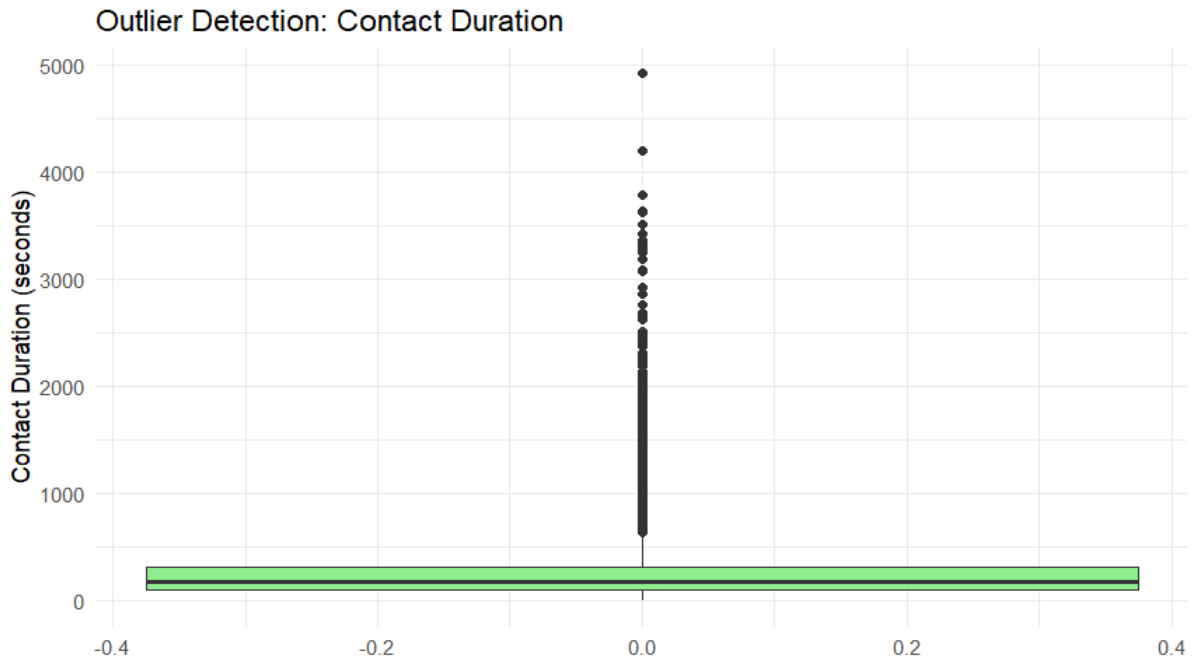


Figure 1 Outlier Detection for Contact Duration

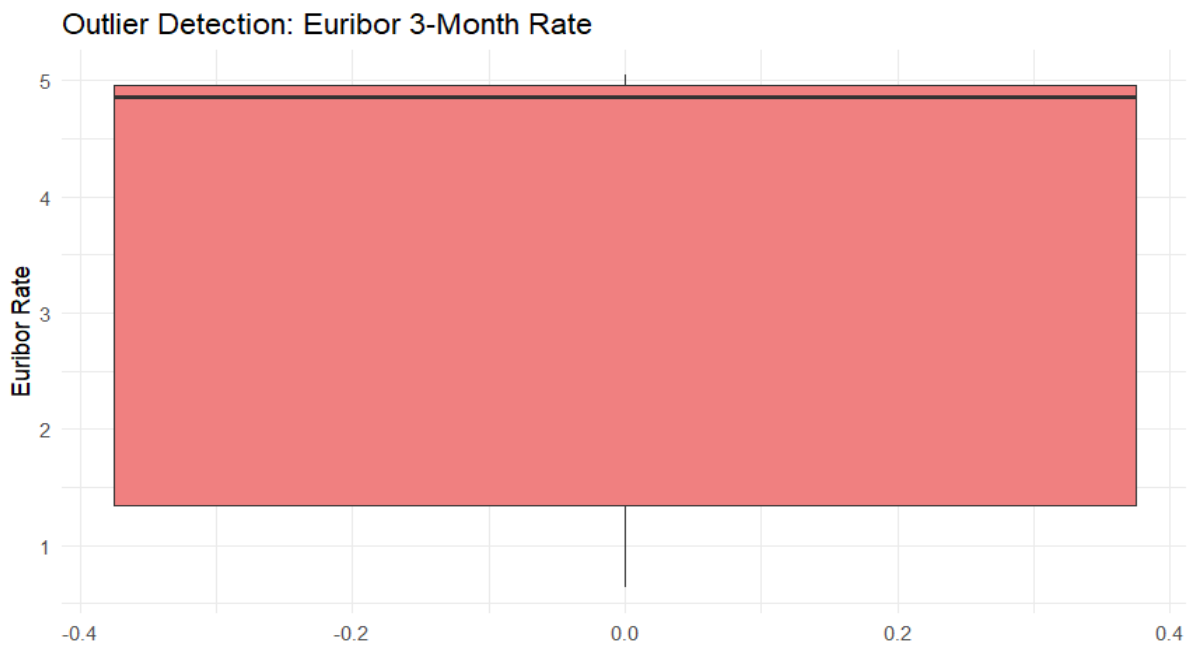


Figure 2 Outlier Detection for Euribor Three-Month Rate

Outlier analysis (shown in the outlier visualisation) identifies extreme values primarily in contact duration. These observations correspond to genuine campaign interactions rather than data errors and were therefore retained to preserve predictive validity and reflect real-world marketing conditions.

3.2 Hypothesis Testing and Exploratory Relationships

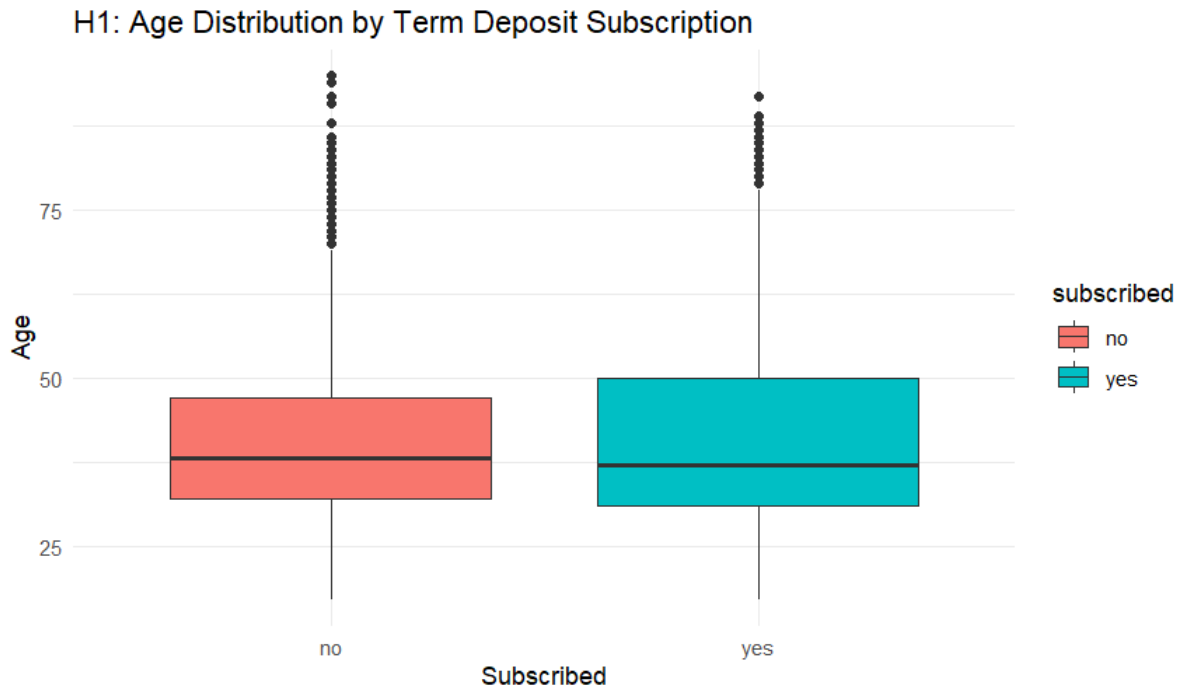


Figure 3 Age Distribution by Term Deposit Subscription Outcome

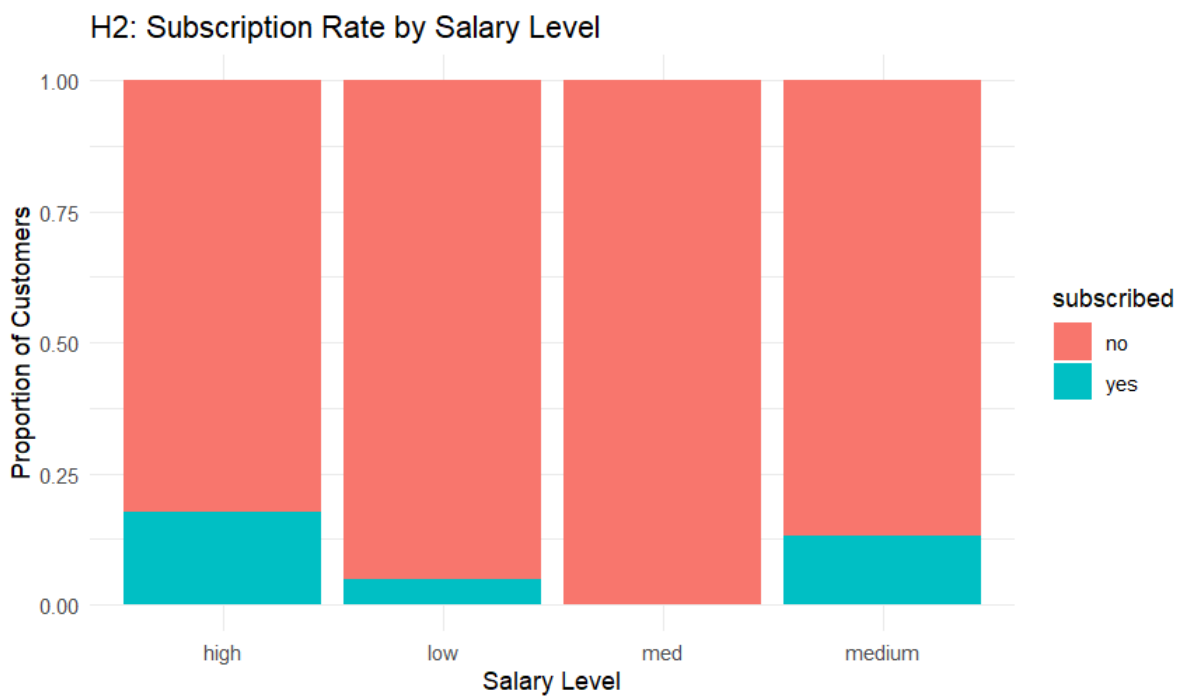


Figure 4 Subscription Proportion by Salary Level

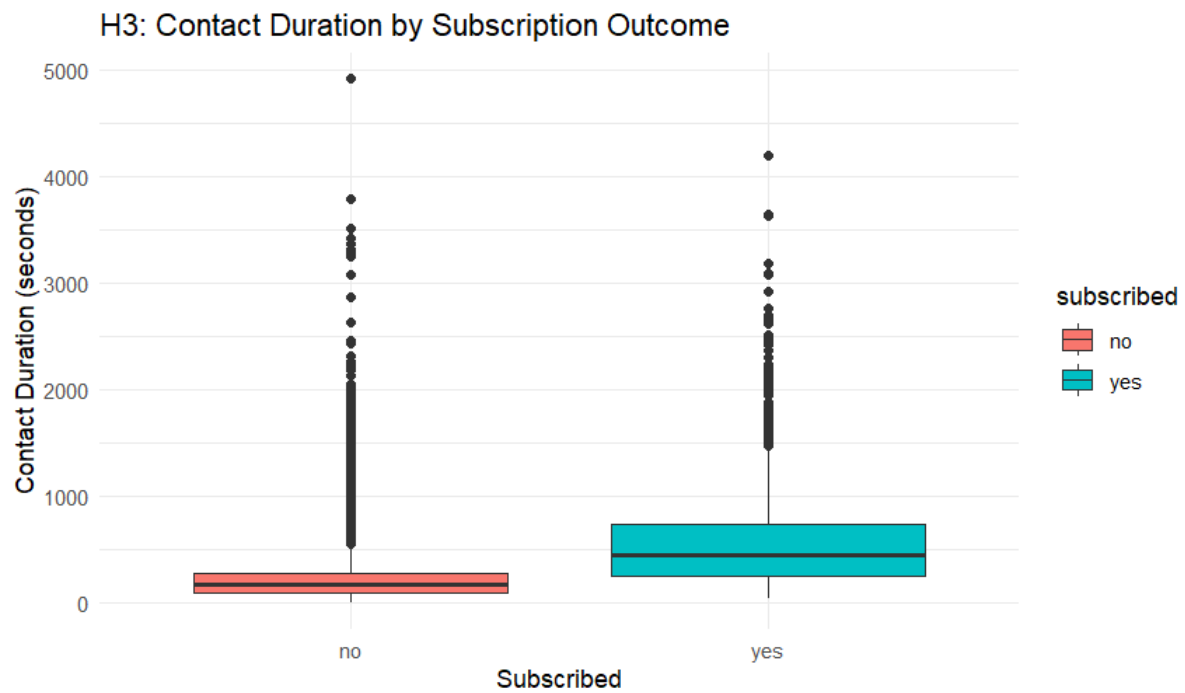


Figure 5 Contact Duration by Subscription Outcome

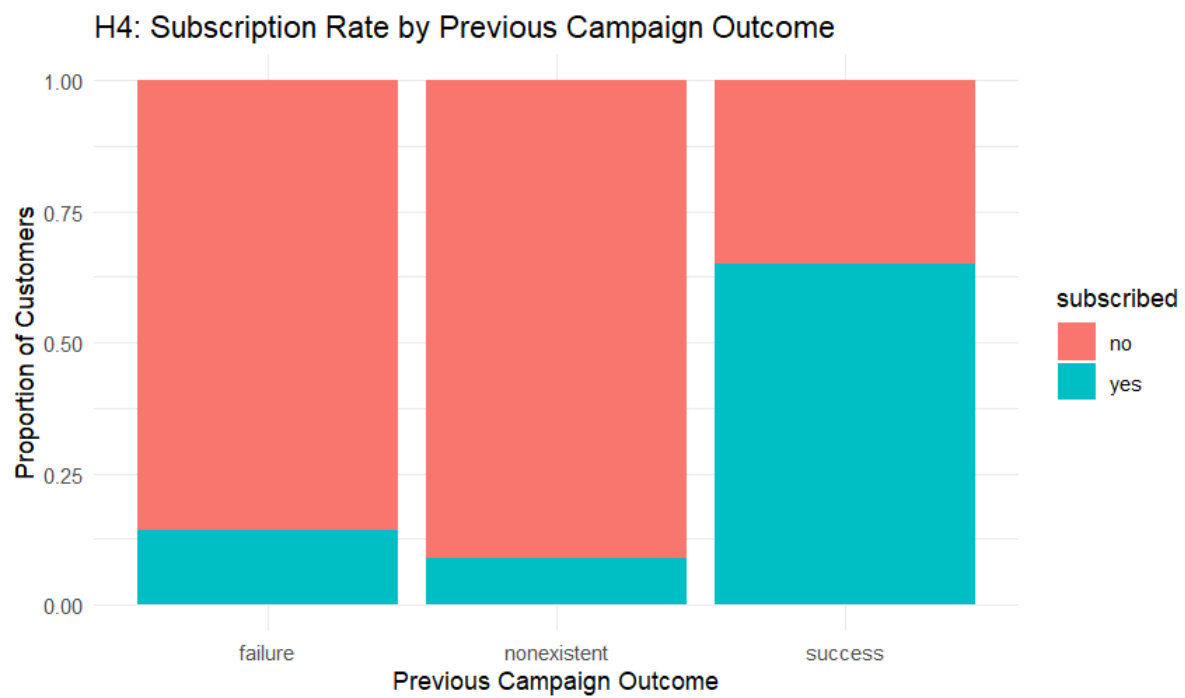


Figure 6 Subscription Rate by Previous Campaign Outcome

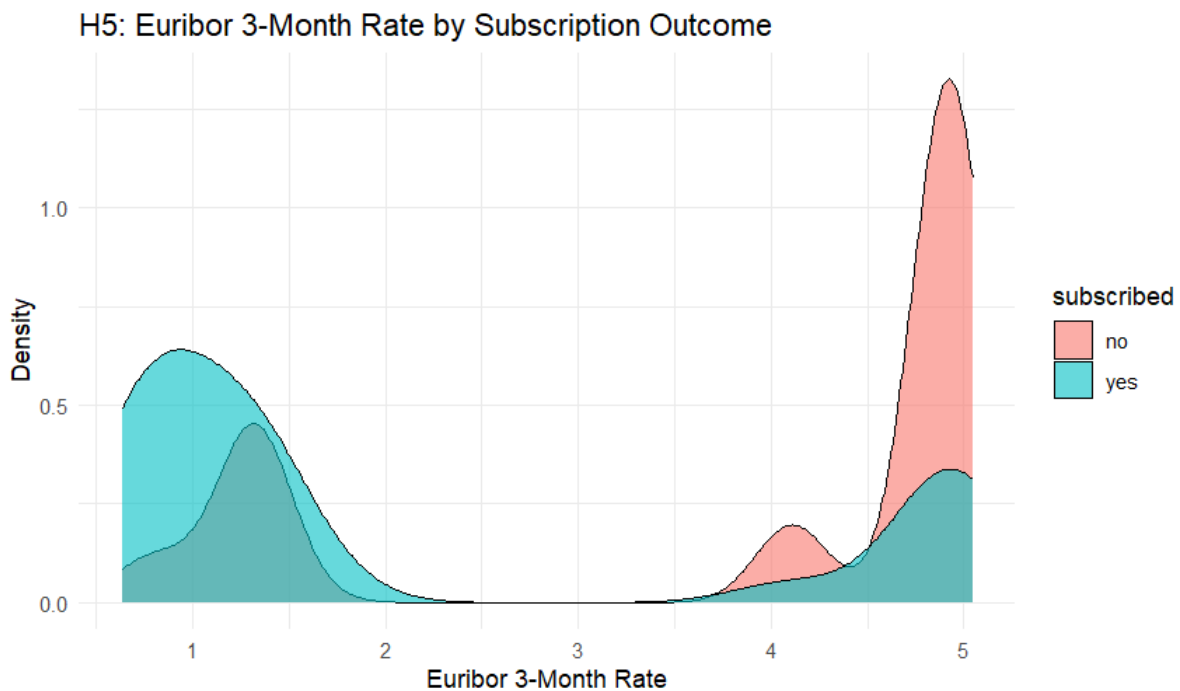


Figure 7 Distribution of Euribor Three-Month Rate by Subscription Outcome

Exploratory analysis provides preliminary support for the study's hypotheses. Customers who subscribed to the term deposit tend to be older, lending support to H1 and aligning with prior evidence that older individuals favour lower-risk financial products. Salary level also demonstrates a meaningful association with subscription behaviour, indicating that income capacity influences customers' willingness to commit funds to fixed-term savings.

Marketing variables display particularly strong relationships with subscription outcomes. Longer contact durations are consistently associated with higher subscription rates, providing strong support for H2 and underscoring the importance of engagement quality during customer interactions. Previous campaign outcomes also show a pronounced effect: customers who previously subscribed exhibit substantially higher conversion rates, offering strong support for H3.

Macroeconomic conditions further differentiate subscribers and non-subscribers. Exploratory visualisation of the Euribor rate suggests that lower interest-rate environments are associated with higher subscription probabilities, supporting H4 and reinforcing the role of broader economic conditions in shaping financial decision-making.

3.3 Logistic Regression Results

# Stargazer output				
> stargazer(model_1, model_2, model_3, type="text",				
+ title="Logistic Regression Models for Term Deposit Subscription",				
+ dep.var.labels="Subscribed (Yes=1)",				
+ covariate.labels=c("Age", "Male", "Married", "Education Level", "Salary Level",				
+ "Contact Method", "Contact Duration", "Number of Contacts",				
+ "Previous Campaign Outcome", "Employment Variation Rate",				
+ "Consumer Price Index", "Consumer Confidence Index",				
+ "Euribor 3-Month Rate", "Number Employed"))				
Logistic Regression Models for Term Deposit Subscription				
=====				
Dependent variable:				

Subscribed (Yes=1)				
	(1)	(2)	(3)	

Age	0.020***	0.018***	0.005**	
	(0.002)	(0.002)	(0.002)	
Male	0.361***	0.356***	0.360***	
	(0.039)	(0.046)	(0.049)	
Married	0.058	-0.019	-0.102	
	(0.065)	(0.076)	(0.081)	
Education Level	0.561***	0.440***	0.120	
	(0.072)	(0.085)	(0.091)	
Salary Level	0.866*	0.561	0.570	
	(0.509)	(0.613)	(0.646)	
Contact Method	0.070	0.006	-0.009	
	(0.077)	(0.092)	(0.099)	
Contact Duration	-0.235***	-0.331***	-0.187*	
	(0.087)	(0.103)	(0.110)	
Number of Contacts	1.266**	1.271*	1.252*	
	(0.626)	(0.712)	(0.740)	
Previous Campaign Outcome	0.181**	0.169*	0.233**	
	(0.083)	(0.099)	(0.108)	
Employment Variation Rate	-0.169	-0.221*	-0.084	
	(0.112)	(0.134)	(0.142)	
Consumer Price Index	0.351***	0.278***	0.263***	
	(0.073)	(0.087)	(0.095)	
Consumer Confidence Index	0.344***	0.326***	0.147	
	(0.105)	(0.126)	(0.137)	
Euribor 3-Month Rate	-1.484***	-1.480***	-1.449***	
	(0.054)	(0.063)	(0.066)	
Number Employed	-11.021	-10.401	-10.043	
	(129.918)	(119.311)	(119.134)	
salary_levelmedium	-0.347***	-0.294***	-0.259***	
	(0.042)	(0.051)	(0.054)	

contact_methodtelephone	-0.989***	-0.899***	
	(0.059)	(0.076)	
contact_duration	0.004***	0.004***	
	(0.0001)	(0.0001)	
number_contacts	-0.111***	-0.049***	
	(0.014)	(0.014)	
prev_campaign_outcomenonexistent	-0.475***	0.427***	
	(0.066)	(0.073)	
prev_campaign_outcomesuccess	2.269***	1.660***	
	(0.095)	(0.102)	
empl_var_rate	-0.812***		
	(0.082)		
cons_price_index	1.126***		
	(0.134)		
cons_conf_index	0.055***		
	(0.008)		
euribor_three_mth	0.028		
	(0.105)		
nr_employed	-0.002		
	(0.002)		
Constant	-2.875***	-3.338***	-96.326***
	(0.139)	(0.179)	(20.681)

Observations	28,658	28,658	28,658
Log Likelihood	-9,445.492	-6,858.243	-5,977.576
Akaike Inf. Crit.	18,922.980	13,758.490	12,007.150
=====			
Note:	*p<0.1; **p<0.05; ***p<0.01		

Three logistic regression models were estimated to assess the combined effects of demographic, marketing, and macroeconomic variables. Model 1, which includes demographic variables only, shows limited explanatory power, with an AIC of 18,922.98. While age and gender are statistically significant predictors, the overall model fit remains modest.

The inclusion of marketing variables in Model 2 leads to a substantial improvement in fit, reducing the AIC to 13,758.49. Contact duration emerges as a highly significant predictor ($p < 0.001$), indicating that longer interactions substantially increase the likelihood of subscription. Previous campaign success also has a strong positive effect, while a higher number of contacts is associated with a reduced probability of subscription, suggesting diminishing returns from repeated outreach.

Model 3 further improves performance by incorporating macroeconomic indicators, achieving the lowest AIC value of 12,007.15, confirming superior model fit. Contact duration remains the strongest predictor across all models, and previous campaign success continues to exert a large, statistically significant effect. Among macroeconomic variables, employment variation rate, consumer price index, and consumer confidence index are significant, while the Euribor rate shows a weaker effect when controlling for other economic factors. These findings collectively support H5, demonstrating the advantage of a multivariate modelling approach.

vif(model_3)			
	GVIF	Df	GVIF^(1/(2*Df))
age	1.483862	1	1.218139
gender	1.003444	1	1.001720
marital_status	1.359010	3	1.052456
education_level	1.193893	7	1.012739
salary_level	1.014494	3	1.002401
contact_method	1.624841	1	1.274692
contact_duration	1.192357	1	1.091951
number_contacts	1.039997	1	1.019802
prev_campaign_outcome	1.322424	2	1.072365
empl_var_rate	32.739613	1	5.721854
cons_price_index	12.875576	1	3.588255
cons_conf_index	3.451258	1	1.857756
euribor_three_mth	61.486946	1	7.841361
nr_employed	42.057204	1	6.485153

Multicollinearity diagnostics indicate no serious violations. Although some macroeconomic variables exhibit higher GVIF values, the adjusted GVIF statistics remain within acceptable limits, supporting the stability of coefficient estimates. Multivariate analysis is essential for capturing the combined effects of demographic, marketing, and macroeconomic variables, as established in prior literature (Hair et al., 2019).

3.4 Predictive Performance and Model Evaluation

confusionMatrix(+ data = test_data\$predicted_class, + reference = test_data\$subscribed +)	
Confusion Matrix and Statistics	
Reference	
Prediction no yes	
no 10652 776	
yes 243 612	
Accuracy : 0.917	
95% CI : (0.912, 0.9219)	
No Information Rate : 0.887	
P-Value [Acc > NIR] : < 2.2e-16	
Kappa : 0.5029	
McNemar's Test P-Value : < 2.2e-16	
Sensitivity : 0.9777	
Specificity : 0.4409	

Pos Pred Value : 0.9321
Neg Pred Value : 0.7158
Prevalence : 0.8870
Detection Rate : 0.8672
Detection Prevalence : 0.9304
Balanced Accuracy : 0.7093
'Positive' Class : no

Predictive performance was assessed using a hold-out test dataset. The confusion matrix shows an overall classification accuracy of 91.7%, exceeding the no-information rate of 88.7%. The model achieves very high sensitivity (97.8%), indicating strong performance in correctly identifying non-subscribers, while specificity (44.1%) is lower. In a marketing context, this imbalance is acceptable, as prioritising potential responders is strategically preferable to missing likely subscribers.

> postResample(+ pred = test_data\$predicted_prob, # predicted probabilities + obs = ifelse(test_data\$subscribed == "yes", 1, 0) # convert factor to numeric +)
RMSE Rsquared MAE
0.2427081 0.4124789 0.1178149

Additional evaluation metrics support these findings. Post-resample results indicate an RMSE of 0.243 and an R² of 0.412, suggesting good generalisation to unseen data.

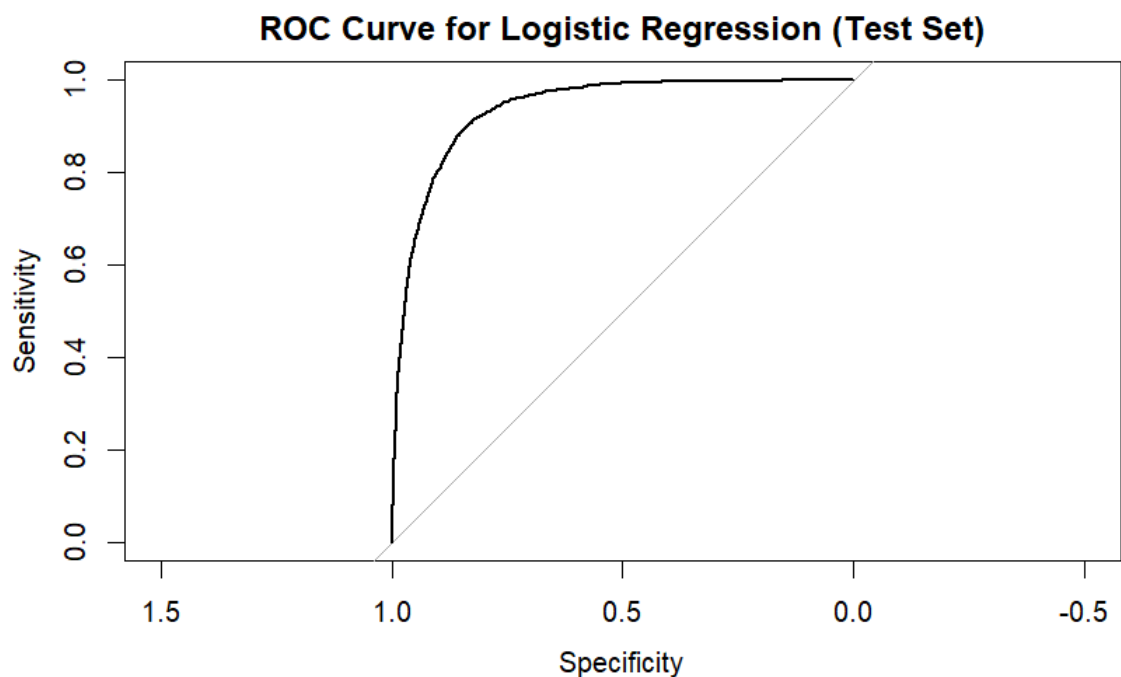


Figure 8 ROC Curve for Logistic Regression Model (Test Set)

> auc(roc_test)
Area under the curve: 0.9366

Evaluation metrics such as ROC curves and AUC are standard measures for assessing predictive performance in business analytics (Provost and Fawcett, 2013)

The ROC curve yields an AUC of 0.9366, indicating excellent discriminative ability.

Overall, the results demonstrate that integrating demographic, campaign-related, and macroeconomic information substantially enhances predictive accuracy and provides actionable insights for improving targeted marketing strategies in the banking sector.

4.0 Conclusion

This study demonstrates that customer subscription to term deposits is strongly influenced by a combination of demographic, campaign-related, and macroeconomic factors. Consistent with prior research, marketing variables—particularly contact duration and previous campaign success—emerge as the most powerful predictors of subscription behaviour, underscoring the importance of interaction quality over repeated contact frequency (Moro, Cortez and Rita, 2014; Santos, Cortez and Rita, 2018). Demographic characteristics such as age also play a meaningful role, aligning with established evidence on risk preferences in financial decision-making (Beckmann and Menkhoff, 2008).

The multivariate logistic regression model incorporating macroeconomic indicators delivers superior predictive performance, confirming the value of integrating broader economic context into customer response modelling (Hosmer, Lemeshow and Sturdivant, 2013; Gujarati, 2020). High classification accuracy and excellent discriminative ability further highlight the model's practical relevance for targeted marketing. Overall, the findings provide actionable insights for banks seeking to optimise campaign efficiency and support data-driven decision-making in competitive financial markets.

5.0 Logbook

Stage 1: Data Exploration and Quality

Date (range): Week 1

Key challenges and decisions: Handling missing values and ensuring the dependent variable (“subscribed”) was correctly formatted for logistic regression. Deciding whether to retain outliers, especially in contact duration, which could bias model estimates.

Converted categorical variables to factors to meet R requirements.

Key findings: Most customers were contacted only a few times. Outliers in contact duration were genuine campaign interactions. Macroeconomic variables were relatively stable, while campaign and demographic variables showed substantial variation.

Stage 2: Data Visualisation

Date (range): Week 1 – Week 2

Key challenges and decision points: Selecting appropriate plots to explore relationships between predictors and subscription outcome. Choosing density plots for continuous variables and bar/histogram plots for categorical variables. Deciding to visualise previous campaign outcomes to assess its effect.

Key findings: Older customers and higher-salary individuals were more likely to subscribe. Longer contact durations and previous positive campaign outcomes were strongly associated with higher subscription rates. Lower Euribor rates coincided with increased subscriptions.

Stage 3: Measures of Association

Date (range): Week 2 – Week 3

Key challenges and decision points: Identifying significant predictors using chi-square tests for categorical variables and t-tests for continuous variables. Checking correlations to assess multicollinearity. Deciding which predictors to include in regression models.

Key findings: Age, salary, contact duration, previous campaign outcome, and certain macroeconomic indicators were significantly associated with subscription. No critical multicollinearity detected among main predictors.

Stage 4: Regression and Prediction

Date (range): Week 3

Key challenges and decision points: Estimating three nested logistic regression models and interpreting coefficients, p-values, and AIC for model selection. Assessing predictive performance with confusion matrices, ROC-AUC, and post-resample metrics. Handling lower specificity in a marketing context.

Key findings: Model 3 (including demographics, campaign, and macroeconomic variables) had the best fit and predictive accuracy. Contact duration and previous campaign success were the strongest predictors. AUC of 0.9366 indicated excellent discriminative ability. Multivariate modelling improved insights compared to simpler models.

6.0 Reflective Commentary

During this module, I have gained substantial experience in applying R for data cleaning, visualization, hypothesis testing, and logistic regression. Initially, I found managing categorical variables and interpreting model outputs challenging, but working systematically through nested models and multicollinearity checks strengthened my confidence. Exploring real-world data highlighted the importance of balancing statistical rigour with practical business considerations, such as accepting lower specificity to prioritise potential marketing responders.

I now feel more confident in developing predictive models and deriving actionable insights from complex datasets. Areas I found particularly interesting included understanding campaign effects on subscription behaviour and integrating macroeconomic indicators into predictive models. Going forward, I plan to apply these skills in financial analytics or marketing roles, using R to support data-driven decisions. I also recognise the need to further improve my proficiency with advanced visualization and automated reporting to streamline future analyses.

7.0 Declaration on Use of Generative AI

I declare that I have not used generative AI tools to produce any part of the substantive content of this assignment. All analyses, interpretations, and narratives have been developed independently using my own knowledge, skills, and academic resources. Any assistance received, including guidance on formatting, referencing, or minor proofreading, has been appropriately acknowledged. I confirm that this submission complies with the University's academic integrity and ethical guidelines.

8.0 References

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Wooldridge, J.M., 2015. *Introductory econometrics: A modern approach*. 6th ed. Boston: Cengage Learning.

Appendix 1: R Code Used

```
## seeding with student number
set.seed(40478904)

## Load packages
install.packages(c("dplyr", "ggplot2", "caret", "corrplot", "car", "lmtest", "stargazer"))
library(dplyr)
library(ggplot2)
library(caret)
library(corrplot)
library(car)
library(lmtest)
library(stargazer)

## Load data and overview
library(readxl)
termsv <- read_excel("C:/Sanjai/byron class/Assignment 2/termsv.xlsx")
View(termsv)

## View structure and summary
str(termsv)
summary(termsv)
head(termsv)
colnames(termsv)

## Data Quality Checks
## Check missing values
colSums(is.na(termsv))

# Remove duplicate rows if any
termsv <- distinct(termsv)

# Check target variable distribution
table(termsv$subscribed)
prop.table(table(termsv$subscribed))

## Convert categorical variables to factors
termsv$gender <- as.factor(termsv$gender)
termsv$occupation <- as.factor(termsv$occupation)
```

```
termsv$salary_level <- as.factor(termsv$salary_level)
termsv$marital_status <- as.factor(termsv$marital_status)
termsv$education_level <- as.factor(termsv$education_level)
termsv$credit_default <- as.factor(termsv$credit_default)
termsv$has_mortgage <- as.factor(termsv$has_mortgage)
termsv$has_personal_loan <- as.factor(termsv$has_personal_loan)
termsv$has_car_insurance <- as.factor(termsv$has_car_insurance)
termsv$has_life_insurance <- as.factor(termsv$has_life_insurance)
termsv$has_savings_account <- as.factor(termsv$has_savings_account)
termsv$has_current_account <- as.factor(termsv$has_current_account)
termsv$has_credit_card <- as.factor(termsv$has_credit_card)
termsv$contact_method <- as.factor(termsv$contact_method)
termsv$month <- as.factor(termsv$month)
termsv$day <- as.factor(termsv$day)
termsv$prev_campaign_outcome <- as.factor(termsv$prev_campaign_outcome)
```

```
## Target variable (important)
```

```
termsv$subscribed <- factor(termsv$subscribed, levels = c("no", "yes"))
```

```
str(termsv)
```

```
## select numeric variables
```

```
numeric_vars <- termsv %>%
```

```
  select(
    age,
    contact_duration,
    number_contacts,
    last_contacted,
    prev_contacts,
    empl_var_rate,
    cons_price_index,
    cons_conf_index,
    euribor_three_mth,
    nr_employed
```

```

)
## Outlier check: Age
ggplot(termsv, aes(y = age)) +
  geom_boxplot(fill = "lightblue") +
  labs(
    title = "Outlier Detection: Age",
    y = "Age"
  ) +
  theme_minimal()

## Outlier check: Contact Duration
ggplot(termsv, aes(y = contact_duration)) +
  geom_boxplot(fill = "lightgreen") +
  labs(
    title = "Outlier Detection: Contact Duration",
    y = "Contact Duration (seconds)"
  ) +
  theme_minimal()

## Outlier check: Euribor Rate
ggplot(termsv, aes(y = euribor_three_mth)) +
  geom_boxplot(fill = "lightcoral") +
  labs(
    title = "Outlier Detection: Euribor 3-Month Rate",
    y = "Euribor Rate"
  ) +
  theme_minimal()

## discriptive statisitcs table
desc_stats <- data.frame(
  Variable = colnames(numeric_vars),
  Mean = sapply(numeric_vars, mean),

```

```

Median = sapply(numeric_vars, median),
SD = sapply(numeric_vars, sd),
Min = sapply(numeric_vars, min),
Max = sapply(numeric_vars, max)
)
desc_stats

## Convert subscribed to factor
termsv$subscribed <- factor(termsv$subscribed, levels = c("no", "yes"))

## Contingency table: Gender vs Subscribed
gender_table <- table(termsv$gender, termsv$subscribed)
gender_table

## Chi-square test
chisq.test(gender_table)

## Salary level vs subscription
salary_table <- table(termsv$salary_level, termsv$subscribed)
salary_table
chisq.test(salary_table)

## Contact method vs subscription
contact_table <- table(termsv$contact_method, termsv$subscribed)
contact_table
chisq.test(contact_table)

## Previous campaign outcome vs subscription
prev_outcome_table <- table(termsv$prev_campaign_outcome, termsv$subscribed)
prev_outcome_table
chisq.test(prev_outcome_table)

## Numeric Variable t test

## Age difference by subscription outcome
t.test(age ~ subscribed, data = termsv)

## Contact duration difference by subscription outcome
t.test(contact_duration ~ subscribed, data = termsv)

## Convert subscribed to numeric for correlation
termsv$subscribed_num <- ifelse(termsv$subscribed == "yes", 1, 0)

```

```

## Correlation with numeric predictors
cor.test(termsv$age, termsv$subscribed_num)
cor.test(termsv$contact_duration, termsv$subscribed_num)
cor.test(termsv$euribor_three_mth, termsv$subscribed_num)

## Hypothesis
## H1: Age vs Subscription
ggplot(termsv, aes(x = subscribed, y = age, fill = subscribed)) +
  geom_boxplot() +
  labs(
    title = "H1: Age Distribution by Term Deposit Subscription",
    x = "Subscribed",
    y = "Age"
  ) +
  theme_minimal()

## H2: Salary Level vs Subscription (Proportions)
ggplot(termsv, aes(x = salary_level, fill = subscribed)) +
  geom_bar(position = "fill") +
  labs(
    title = "H2: Subscription Rate by Salary Level",
    x = "Salary Level",
    y = "Proportion of Customers"
  ) +
  theme_minimal()

## H3: Contact Duration vs Subscription
ggplot(termsv, aes(x = subscribed, y = contact_duration, fill = subscribed)) +
  geom_boxplot() +
  labs(
    title = "H3: Contact Duration by Subscription Outcome",
    x = "Subscribed",
    y = "Contact Duration (seconds)"
  ) +
  theme_minimal()

```

```
## H4: Previous Campaign Outcome vs Subscription
```

```
ggplot(termsv, aes(x = prev_campaign_outcome, fill = subscribed)) +  
  geom_bar(position = "fill") +  
  labs(  
    title = "H4: Subscription Rate by Previous Campaign Outcome",  
    x = "Previous Campaign Outcome",  
    y = "Proportion of Customers"  
  ) +  
  theme_minimal()
```

```
## H5: Euribor Rate vs Subscription
```

```
ggplot(termsv, aes(x = euribor_three_mth, fill = subscribed)) +  
  geom_density(alpha = 0.6) +  
  labs(  
    title = "H5: Euribor 3-Month Rate by Subscription Outcome",  
    x = "Euribor 3-Month Rate",  
    y = "Density"  
  ) +  
  theme_minimal()
```

```
## test and train set split
```

```
library(caret)
```

```
# 70% train, 30% test split
```

```
train_index <- createDataPartition(termsv$subscribed, p = 0.7, list = FALSE)
```

```
train_data <- termsv[train_index, ]
```

```
test_data <- termsv[-train_index, ]
```

```
# Check distribution of target variable in train/test
```

```
prop.table(table(train_data$subscribed))
```

```
prop.table(table(test_data$subscribed))
```

```
## Logistic Regression Models
```

```
# Model 1: Demographics only
```

```
model_1 <- glm(subscribed ~ age + gender + marital_status + education_level + salary_level,  
               data = train_data, family = binomial(link = "logit"))
```

```
summary(model_1)
```

```

# Model 2: Add marketing variables
model_2 <- glm(subscribed ~ age + gender + marital_status + education_level + salary_level +
               contact_method + contact_duration + number_contacts + prev_campaign_outcome,
               data = train_data, family = binomial(link = "logit"))
summary(model_2)

# Model 3: Add macroeconomic indicators
model_3 <- glm(subscribed ~ age + gender + marital_status + education_level + salary_level +
               contact_method + contact_duration + number_contacts + prev_campaign_outcome +
               empl_var_rate + cons_price_index + cons_conf_index + euribor_three_mth +
               nr_employed,
               data = train_data, family = binomial(link = "logit"))
summary(model_3)

# Compare model fit using AIC
AIC(model_1, model_2, model_3)

# Stargazer output
stargazer(model_1, model_2, model_3, type="text",
           title="Logistic Regression Models for Term Deposit Subscription",
           dep.var.labels="Subscribed (Yes=1)",
           covariate.labels=c("Age","Male","Married","Education Level","Salary Level",
                              "Contact Method","Contact Duration","Number of Contacts",
                              "Previous Campaign Outcome","Employment Variation Rate",
                              "Consumer Price Index","Consumer Confidence Index",
                              "Euribor 3-Month Rate","Number Employed"))

# Check multicollinearity
vif(model_3)

# Predict probabilities on test set
test_data$predicted_prob <- predict(
  model_3,
  newdata = test_data,
  type = "response"
)

```

```

)
# Check if any NA values exist
sum(is.na(test_data$predicted_prob))
# Classification is done manually instead of using ifelse()
# to avoid session crashes and to handle any potential NA values safely.
# Create predicted class using a safe approach
test_data$predicted_class <- NA
test_data$predicted_class[test_data$predicted_prob >= 0.5] <- "yes"
test_data$predicted_class[test_data$predicted_prob < 0.5] <- "no"
# Convert to factor
test_data$predicted_class <- factor(
  test_data$predicted_class,
  levels = c("no", "yes")
)
# Confusion matrix on test set
confusionMatrix(
  data = test_data$predicted_class,
  reference = test_data$subscribed
)
## Post-resample model performance on test set
## This provides Accuracy, Kappa, and (for classification via probabilities)
## compares predicted probabilities against the true class labels.
## Used as an additional evaluation metric after the confusion matrix.
postResample(
  pred = test_data$predicted_prob,          # predicted probabilities
  obs = ifelse(test_data$subscribed == "yes", 1, 0) # convert factor to numeric
)
## ROC Curve & AUC on test set
library(pROC)
roc_test <- roc(test_data$subscribed, test_data$predicted_prob)
plot(roc_test, main="ROC Curve for Logistic Regression (Test Set)")
auc(roc_test)

```